Jinming Xing

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Education

North Carolina State University **Sep 2023 – May 2027**

- PhD in Computer Science
- **Research Interest:** Graph Neural Networks, Large Language Models

Shenzhen University **Sep 2019 – Jun 2023**

- BS in Computer Science (Honored)
- **Coursework:** Probabilities, Linear Algebra, Data Structures and Algorithms, Computer Networks, Internet of Things, Cloud Computing, Database, Machine Learning, Practical Deep Learning, Computer Vision

Skills

- Python, C/C++, PyTorch, Hadoop, Spark, MySQL, Git, Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn
- Agent, LLM (LlamaIndex, Ollama, RAG, LoRA/PEFT, MCP, GRPO), Transformers, CNNs, RNNs, GNNs

Industry Experience

MLE Intern **Adobe** **May 2026 – Aug 2026**

Predictive Digital Twin: Counterfactual Simulation of Marketing Journeys on Adobe Experience Platform (lead inventor, patent approved by Adobe review committee, in drafting with counsel; research paper in preparation)

[Motivation]: A customer's response to a campaign they never received is a counterfactual that historical logs, confounded by past targeting, cannot reveal, and A/B tests to find it are slow, costly, and risky.

- Built a treatment-conditioned sequence model (GNN world model, gated-recurrent latent-state supernode, per-action heads, g-computation rollout) that simulates a customer's trajectory under any hypothetical campaign, deconfounds effects that historical logs can't reveal, and answers casual what-if questions.
- Authored a controllable synthetic benchmark with exact counterfactuals; improved individual treatment-effect accuracy by 67-70% over baselines, beating gradient boosting and the Causal/G-Transformer by 36-46 points.
- Validated on real anonymized adobe.com clickstream (matches the performance of the Adobe deployed model in churn prediction and outperforms it in several tasks) and public OPE/uplift datasets (Open Bandit, Criteo); shipped a Streamlit demo and an Adobe CoWorker MCP for marketing campaign optimization and targeting.

Selected Papers & Research Experience

Research Assistant **North Carolina State University** **Oct 2023 – May 2027**

Jinming Xing, Muhammad Shahzad, et al. "When Text and Topology Disagree: Bidirectional GNN-LLM Co-Evolution for Noisy Graph Learning." (WSDM'27, Under Review)

[Motivation]: Existing LLM-GNN integration relies on static, unidirectional pipelines, causing bidirectional error propagation, semantic-structural dissonance, and blind alignment that propagates noise.

- Introduce a Dual-View Co-Evolution mechanism where the GNN injects topological context to guide the LLM, while the LLM constructs a Dynamic Semantic Graph to rewire the GNN, halting mutual error propagation.
- Propose a Node-Adaptive Gating Mechanism that learns per-node factors to dynamically fuse static and semantic graphs, resolving the missing link problem by reconstructing connections for isolated nodes.
- Design a Hard-Structure Conflict-Aware Contrastive Loss based on Global Graph Diffusion (PPR), targeting false semantic friends by warping the semantic manifold to respect high-order topological boundaries.
- Develop an Uncertainty-Gated Consistency strategy that enables meta-cognitive alignment, ensuring models only learn from the confident counter-view, and an Entropy-Aware Adaptive Fusion for robust inference.
- Outperforming SOTA LLM-GNN integration baselines by up to 9.07% and 7.19% in accuracy and F1-score.

Jinming Xing, Guoheng Sun, et al. "NetSight: Joint Spatio-Temporal Dependency Modeling for Network Traffic Prediction." (CIKM'26)

[Motivation]: Hidden ST dependencies are ignored. Global-local ST can't be modeled simultaneously.

- A dynamic data-driven spatial-temporal graph was proposed to capture hidden spatial-temporal dependencies.
- A decoder-only transformer was adopted while GAT servers as the graph convolution backbone.
- Proposed NetSight, which copes with global-local spatial-temporal patterns simultaneously. A simple yet effective learnable pooling was designed, and to speed up the convergence, node normalization was introduced.
- NetSight outperforms the average of baselines by 36.37% (MAE), 32.19% (RMSE), and 23.99% (SMAPE).

Jinming Xing, Muhammad Shahzad. ”A Reinforcement Learning Framework for Application-Specific TCP Congestion-Control.” (IPCCC’25)

[Motivation]: Diverse optimization goals are overlooked. Client computational load is high.

- Proposed ASC_{RL}, a DRL-based framework that can accommodate clients with various optimization goals.
- Designed a client-server updating mechanism, which decouples the training and inference, reducing the client computational load by up to 91%, making it suitable to be deployed on low-power devices.
- Introduced a fast retraining mechanism for client goals changing, resulting in a 50% reduced retraining time.

Research Assistant | **Shenzhen University** | **Jun 2021 – Jul 2023**

Jinming Xing, Can Gao, et al. ”Weighted fuzzy rough sets-based tri-training and its application to medical diagnosis.” Applied Soft Computing [Q1, IF:7.28] (2022)

[Motivation]: Noisy data impedes the learning. Tri-training suffers from the initialization problem.

- Proposed the ‘bad-point’ technique for dataset de-noising and a high-order information extraction strategy.
- Three modal data ‘ORI’, ‘PCA’, and ‘DIS’ are proposed to initialize tri-training base classifiers.
- Designed a robust weighted fuzzy lower approximation classifier for supervised and semi-supervised problems.
- The proposed WFRS-Tri framework outperforms baselines under various noisy supervised and semi-supervised scenarios with a range of accuracy improvements from 3% to 15% on average.

Grants

- North Carolina State University College of Engineering Travel Grant, 2025, **$1k**
- National College Students Innovation and Entrepreneurship Training Project, 2021, **$2.8k**
- Guangdong Province Science and Technology Innovation Strategy Special Fund Project, 2022, **$2.1k**

Services

Invited Seminars: ”Introduction to GNNs and Their Applications”, Jharkhand Rai University, India, May 2025

Teaching Assistant: Computer Network, Data Structures and Algorithms, Senior Design

Program Committee: AAAI’27, WWW’25 (Short Paper Track)

Reviewer: NeurIPS’26, AAAI’26, ICDM’26, WWW’25 (Main Track), TMM, TKDD, IJCNN’25/26, Soft Computing, Knowledge-Based Systems, Expert Systems With Applications, IJAR, Applied Soft Computing, Neruo-Computing, The Journal of Supercomputing, Results in Engineering, Advanced Engineering Informatics, Computer Communications

Full List of Peer-Reviewed Publications

- **Jinming Xing**, Guoheng Sun, Can Gao, et al. ”Looking for Bidding Teammates: A Game-Theoretic Model of Stranger Collusion in Peer Review.” [ICLR’27, Under Review]
- **Jinming Xing**, Charlotte Brian. ”Governance Outruns Analysis: Incidence and Manipulability of Credential-Gated Reciprocal Review.” [NeurIPS Workshop’26, Under Review]
- **Jinming Xing**, Muhammad Shahzad. ”When Text and Topology Disagree: Bidirectional GNN-LLM Co-Evolution for Noisy Graph Learning.” [WSDM’26, Under Review]
- Chang Xue, Fang Liu, Jiaye Wang, **Jinming Xing**, Chen Yang. ”TAS-GNN: A Status-Aware Signed Graph Neural Network for Anomaly Detection in Bitcoin Trust Systems.” ICPR’26.
- Chang Xue, Youwei Lu, Chen Yang, **Jinming Xing**. ”RecMind: LLM-Enhanced Graph Neural Networks for Personalized Consumer Recommendations.” ICCE’26. IEEE, 2026. [EI indexed]
- Guoheng Sun, Ziyao Wang, Xuandong Zhao, Bowei Tian, Zheyu Shen, Yexiao He, **Jinming Xing**, Ang Li. ”Invisible Tokens, Visible Bills: The Urgent Need to Audit Hidden Operations in Opaque LLM Services.” ICML’26.
- **Jinming Xing**, Guoheng Sun, Hui Sun, et al. ”NetSight: Joint Spatio-Temporal Dependency Modeling for Network Traffic Prediction.” CIKM’26.
- **Jinming Xing**, Muhammad Shahzad. ”A Reinforcement Learning Framework for Application-Specific TCP Congestion-Control.” IPCCC’25, IEEE, 2025.
- Qisen Cheng, **Jinming Xing**, Chang Xue, et al. ”Unifying Prediction and Explanation in Time-Series Transformers via Shapley-based Pretraining.” CSPA’25. IEEE, 2025. [EI indexed]
- **Jinming Xing**, Dongwen Luo, Qisen Cheng, et al. ”Multi-view Fuzzy Graph Attention Networks for Enhanced Graph Learning.” IWPR’25. 2025. [EI indexed]
- **Jinming Xing**, Chang Xue, Dongwen Luo, et al. ”FGATT: A Robust Framework for Wireless Data Imputation Using Fuzzy Graph Attention Networks and Transformer Encoders.” ECCS’25. IEEE, 2025. [EI indexed]

- **Jinming Xing**, Zhaomin Xiao, Yingyi Wu, et al. ”Network Traffic Forecasting via Fuzzy Spatial-Temporal Fusion Graph Neural Networks.” ISCMI’24, pp. 282-286. IEEE, 2024. [EI indexed]
- **Jinming Xing**, Qisen Cheng, Dongwen Luo, et al. ”Comparative Analysis of Pooling Mechanisms in LLMs: A Sentiment Analysis Perspective.” ISMSI’25. ACM, 2025. [EI indexed]
- **Jinming Xing**, Ruilin Xing, Qisen Cheng, et al. ”Enhancing Link Prediction with Fuzzy Graph Attention Networks and Dynamic Negative Sampling.” ICMSSP’25. Springer Nature, 2025. [EI indexed]
- Can Gao, Jie Zhou, **Jinming Xing**, et al. ”Parameterized maximum-entropy-based three-way approximate attribute reduction.” International Journal of Approximate Reasoning 151 (2022): 85-100. [Q2, IF: 3.0]
- **Jinming Xing**, Can Gao, et al. ”Weighted fuzzy rough sets-based tri-training and its application to medical diagnosis.” Applied Soft Computing 128 (2022): 109467. [Q1, IF:6.6]